TECHNICAL PRESENTATION

Big Data Analytics in Cloud Computing

Team No. [12]

CSA15 – Cloud Computing and Big Data Analytics

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Abstract

- **Big data:** huge volume, velocity, variety
- **Problem:** on-premise systems are costly and hard to scale
- **Cloud:** elastic storage, compute, managed analytics
- **Architecture:** data lake, batch + stream, warehouse
- **Platforms:** AWS, Azure, Google Cloud
- **Value:** faster insight, lower cost, better decisions
- **Case study:** Netflix on AWS

1 Architectures
Cloud data-processing layers

2 Platforms
AWS • Azure • Google Cloud

3 Value & Use Cases
Business impact, Netflix case

Aim and Objective

AIM
Understand how cloud computing enables scalable, cost-effective big data analytics

1. **Explore architectures**
   Cloud data-processing layers and pipelines
2. **Compare platforms**
   AWS, Azure and Google Cloud analytics services
3. **Assess business value**
   Cost, speed, scale and decision quality
4. **Study real applications**
   Netflix case study on AWS

Introduction

- **Big data:** datasets too large or fast for traditional tools
- **Sources:** IoT, social media, logs, transactions
- **Cloud computing:** on-demand compute and storage [3]
- **Pay-as-you-go,** elastic, globally available
- **Cloud + big data:** analytics without owning hardware

The 5 Vs of Big Data

Volume
TB to petabyte scale

Velocity
Real-time data streams

Variety
Structured to unstructured

Veracity
Data quality and trust

Value
Insight that drives decisions

Core Technical Explanation

PROBLEM STATEMENT

Traditional on-premise analytics cannot economically store and process growing, fast, diverse data

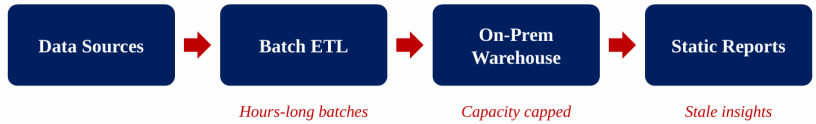
Scalability	Cost	Latency	Governance
Fixed hardware, long procurement, peak overload	High up-front CapEx, idle capacity most of the time	Batch-only jobs deliver late insights	Data silos, security and compliance gaps

Research need: *elastic, secure, unified analytics on the cloud [1]*

Existing System

- **On-premise** Hadoop clusters and enterprise data warehouses
- **Fixed capacity** bought up-front (high CapEx)
- **Batch ETL:** insights arrive hours or days late
- **Hard to scale** for peaks; idle in quiet periods
- **Siloed data** and heavy manual maintenance

Typical on-premise pipeline



Proposed System

Cloud-native analytics platform: managed, elastic, pay-per-use

1 Elastic Storage

Data lake on object storage; scales to petabytes

2 Managed Processing

Batch and stream engines without cluster upkeep

3 Serverless Warehouse

SQL analytics; pay only for queries run

4 Self-Service BI & ML

Dashboards, predictions and AI services

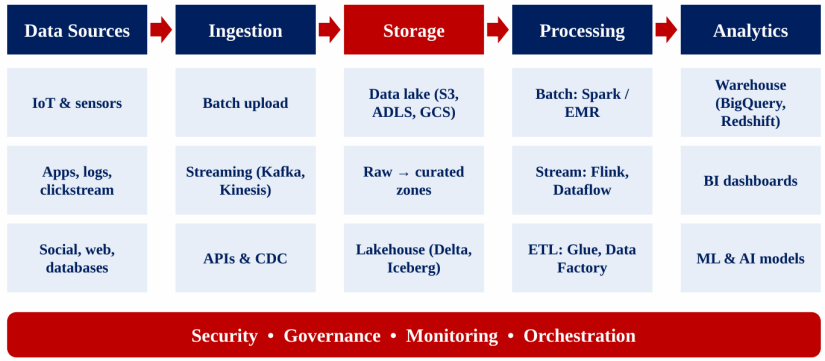
5 Security & Governance

Encryption, access control, audit trails

6 Pay-as-You-Go

Scale up for peaks, scale down to save

Overall System Architectural Design



Layered, loosely coupled design: every layer scales independently

Proposed Methodology and Algorithm

Analytics pipeline: raw data to decision

1 Ingest → 2 Store → 3 Clean & Transform → 4 Analyze → 5 Visualize & Act

Algorithm: MapReduce [4] on Apache Spark [5]

Input Splits	Map	Shuffle & Sort	Reduce	Output
Data blocks across workers	Process partitions in parallel	Group records by key	Aggregate each key	Results to cloud storage

- **Parallel:** workers process data blocks simultaneously
- **Fault tolerant:** failed tasks re-run automatically
- **Elastic:** add nodes on demand, release when done

Cloud Analytics Platforms

Layer	AWS	Microsoft Azure	Google Cloud
Storage	Amazon S3	ADLS Gen2	Cloud Storage
Batch / ETL	EMR, Glue	Databricks, Data Factory	Dataproc, Dataflow
Streaming	Kinesis, MSK	Event Hubs	Pub/Sub, Dataflow
Warehouse	Redshift	Synapse Analytics	BigQuery
BI & ML	QuickSight, SageMaker	Power BI, Azure ML	Looker, Vertex AI

Choice depends on existing stack, skills, pricing and compliance needs

Business Value

Lower Cost
- No up-front hardware (CapEx to OpEx)
- Pay only for what you use

Faster Insights
- Real-time streams, self-service BI
- Shorter time from data to decision

Scale & Agility
- Petabytes on demand, global reach
- Launch new analytics quickly

Smarter Decisions
- Personalization, fraud detection, forecasting
- ML and AI services built in

ROI = faster decisions + lower infrastructure cost + data-driven products

Real-world Use Case: Netflix on AWS

Netflix Keystone: Kafka-based event pipeline on AWS [2]

~500 B	~1.3 PB	~8 M
events per day (2016)	data per day	events per second at peak

How the data flows

Apps & Devices (Playback, UI events) → Kafka (Buffers events) → Routing Service (Fans out data) → S3 / Elasticsearch (Store and search) → Analytics (Hive, Spark, EMR)

- **Insights drive** recommendations, playback quality and operations
- **Volume has grown** past 1 trillion events per day since 2016

Conclusion

Problem	Solution	Impact
• Legacy analytics: costly, rigid, slow • Data growing in volume, speed, variety	• Cloud data lake with batch + stream processing • Managed platforms: AWS, Azure, GCP	• Elastic scale and lower cost • Faster, data-driven decisions • Proven at Netflix scale

VALUE & SIGNIFICANCE

Cloud makes big data analytics scalable, affordable and accessible

References

[1] I. A. T. Hashem et al., "The rise of 'big data' on cloud computing: Review and open research issues," Information Systems, vol. 47, pp. 98–115, 2015.

[2] Netflix Technology Blog, "Evolution of the Netflix Data Pipeline," Feb. 2016. techblog.netflix.com/2016/02/evolution-of-netflix-data-pipeline.html

[3] M. Armbrust et al., "A View of Cloud Computing," Communications of the ACM, vol. 53, no. 4, pp. 50–58, 2010.

[4] J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters," Communications of the ACM, vol. 51, no. 1, pp. 107–113, 2008.

[5] M. Zaharia et al., "Apache Spark: A Unified Engine for Big Data Processing," Communications of the ACM, vol. 59, no. 11, pp. 56–65, 2016.

[6] AWS, Microsoft Azure and Google Cloud, analytics services product documentation.